

Scientific Reviewer

WEIGHT 4 · substantial

Claim · strengths · biting ablations · weaknesses · ACT TENTATIVE · accept-level
RSS systems

Verdict preview

Accept-level RSS systems + data-interface paper. Empirics real-robot and ablation-rich. Punish overgeneralizing from cup-centric OOD and SLAM-friendly scenes.

Claim under review

Zero-shot transferable dynamic / bimanual / long-horizon from wild human demos

CLAIM

Careful handheld interface + latency-matched relative-trajectory Diffusion Policy enables zero-shot transferable dynamic, bimanual, and long-horizon skills from in-the-wild human demonstrations — without robots during collection.

Spine check: relative SE(3) · latency matching · wrist fisheye · DP $T_a=6$ · open stack

Strengths

Systems completeness · biting ablations · cross-emb · OOD lesson · open stack

Systems completeness Closes four prior failure modes: FoV, SfM precision, latency, multimodal capacity — each with concrete design.

Ablations that bite Fisheye, mirrors+reflect, rel vs abs/delta, latency, inter-gripper, ViT vs ResNet — large effect sizes, matched inits.

Cross-embodiment Same cup ckpt UR5→Franka 90% with honest joint-limit fails (2/20).

OOD data lesson ViT-L + narrow data = 0%; diversity required — antidote to “just finetune CLIP.”

Open hardware+software MIT stack sufficient for third-party reproduce (SLAM hard — disclosed).

Builds on DP Correctly uses Diffusion Policy rather than inventing a new generative head.

Fig. 8 — ablation bars that bite

Exact endpoints: abs 25% · delta 80% · no fisheye 55% · no lat 57.5% · cloth 30% · ResNet 0%



Prefer reporting fractions (20/20, 105/120, 43/60) over vague “~70%”

Cross-embodiment + OOD lesson

UR5 100% → Franka 90% · wild 71.7% vs narrow 0%

100%

UR5 cup (same ckpt)

90%

Franka FR2 (2× JL)

71.7%

wild OOD 43/60

0%

narrow ViT OOD

- Cross-emb is a FEATURE of relative EE + matched wrist geometry — not a free lunch (kin-filter drops demos).
- OOD: 1400 demos / 30 locations / 12 person-hours beats pretrained ViT on narrow data alone.
- Report 43/60 precisely — “70% OOD” blends train/test cups and two scenes.
- Honest residual: Franka fails are joint-limit, not policy confusion.

Weaknesses (1/2)

n=20 · SLAM texture · kin-filter post-hoc · 780 g

Small eval n

Often n=20 episodes; operator-judged success with subjective rubrics — videos help but stats thin.

SLAM texture dependence

Admitted limitation; wild success conditioned on mappable scenes.

Kinematic filtering post-hoc

Pushes embodiment limits to data drop, not embodiment-aware learning.

780 g bulk/weight

Still slower than bare hand; limits marathon collection sessions.

Weaknesses (2/2)

Parallel-jaw · 100/120 ms examples · report 43/60 not vague 70%

Parallel-jaw only

No dexterous-hand transfer story — ceiling on contact-rich tasks.

Latency examples $\neq$ constants

Fig. 5's 100 ms arm / 120 ms gripper are per-install examples — measure locally.

Headline rounding

Prefer 43/60 = 71.7% over vague “~70% OOD.” Break out cafe vs fountain when claiming.

ACT drop-in TENTATIVE

Paper says ACT could drop in — NOT empirically compared. Do not assert equal.

Methodological notes

Matched inits · absolute strawman? · ACT TENTATIVE

Fair matched initial-state protocol is a plus; timeout/safety termination subjectivity remains.

Absolute-action baseline may look like a strawman under perfect calib — but paper's point is perfect calib is unrealistic in wild, so relative wins on robustness.

ACT-as-drop-in claimed but not ablated — mark TENTATIVE if asserted as equal empirical swap.

Main risk: overgeneralizing from cup-centric OOD and SLAM-friendly scenes.

VERDICT

Accept-level RSS systems contribution

ACCEPT as landmark systems + data-interface paper

Empirics are real-robot and ablation-rich. Interface knobs (relative, latency, fisheye, mirrors) are the contribution — not a new generative head.

Punish: overclaiming “generalist” from cup OOD; treating 100/120 ms as universal; equating ACT drop-in with measured parity; inventing GPU hours.

Exam-ready one-liner

UMI = handheld wild teaching without robots; biting ablations (rel 100 vs abs 25; lat 87.5→57.5; wild 71.7 vs 0); accept systems, punish overclaim.

Fig. 9 — wild OOD cafe / fountain

Evidence for diversity > ViT alone · report 43/60

Narrow-domain + same ViT family → 0% in those scenes



Numbers tattoo — pin these

Every deck carries the same spine numbers

100% / 90%

cup UR5 / Franka

87.5%

toss w/ latency

57.5%

toss no latency

71.7%

wild 43/60 OOD

0%

narrow → wild OOD

100% vs 25%

rel vs abs action

55%

no fisheye (69°)

70% / 30%

cloth ± inter-grip

70% / 0%

dish ViT / ResNet

1400 / 30

wild demos / locs

Ta=6

@10Hz (20Hz toss)

73 + 298

BOM · 780g

Exam bite — Reviewer

What to write under time pressure

Accept systems interface paper; cite biting ablations (rel/abs/delta, latency, fisheye, wild vs narrow); flag $n=20$, SLAM texture, kin-filter post-hoc, parallel-jaw ceiling; mark ACT equal empirical as TENTATIVE; report 43/60 not vague 70%.

Scientific Reviewer — close

Takeaways + exam bite

1. Strengths: systems completeness + biting ablations + cross-emb + OOD data lesson.
2. Weaknesses: n=20, SLAM texture, kin-filter post-hoc, 780g, jaw-only, latency examples.
3. ACT drop-in = TENTATIVE; report 43/60 precisely.
4. Verdict: accept-level RSS systems; punish generalist overclaim.

EXAM BITE

Accept UMI as systems+interface; punish overclaim; ACT equal = TENTATIVE.